Retention Strategy Memo - US Grocery Loyalty Panel

To: Commercial leadership | **From:** Customer Analytics | **Scoring date:** 2017-10-01 (back-test) **Basis:** Complete Journey 2.0 (84.51, CC0), 2,469 loyalty-card households, 2017-01-01 to 2017-12-31. Every figure is computed by the pipeline in this repository. Items marked **ASSUMED** are scenario inputs, not observed results. **Churn is a proxy**: no purchase in the 90 days after the cutoff, not an observed cancellation.

1. The decision

Which active households should receive retention spend (coupons, outreach) next quarter, and how much is their future spend plausibly worth?

2. What the data says (observed)

- 2,469 households made 155,368 clean baskets worth $4.60M across 457 stores. The top 10% of households account for 35% of spend.
- Store discounts are 14.7% of shelf value and coupon discounts only 0.57%; 6.3% of baskets use a coupon.
- Households rarely go quiet: 93.4% remain lapse-free at day 90 (Kaplan-Meier, landmark design). Baseline trip frequency separates them sharply: day-90 survival is 86.8% for the lowest trip tier and 99.7% for the highest (Cox hazard ratio per log trip 0.25, 95% CI 0.19 to 0.34; the proportional-hazards test flags this effect as time-varying).

3. Churn-proxy model (out-of-time test, 2,314 households, 89 positives)

- Prevalence is 3.8% (train 3.3%, validation 4.4%).
- Test ROC-AUC with 95% bootstrap intervals: recency rule 0.793 (0.741 to 0.847), RFM heuristic 0.806, logistic regression 0.841 (0.803 to 0.875), LightGBM 0.843 (0.804 to 0.878). PR-AUC: 0.144, 0.123, 0.152, 0.161 against a 0.038 base rate.
- **LightGBM is not distinguishable from logistic regression**: paired-bootstrap ROC-AUC difference +0.002 (-0.012 to +0.017). Logistic regression beats the RFM heuristic by +0.034 (+0.015 to +0.055). In the rolling backtest LightGBM matches or beats logistic regression at 5 of 5 cutoffs.
- Final model: LightGBM + platt (best validation PR-AUC); top-decile lift 4.03x. **Calibration did not improve Brier on test**: raw LightGBM 0.0345, final 0.0355. Mean predicted risk is 3.1% raw and 4.4% after Platt scaling versus 3.8% observed; calibration slope 0.78. Isotonic was ineligible with 100 validation positives. Use scores for ranking.
- Main drivers (mean |SHAP|): trips per week, trips 42d, trips 28d, share fresh.

4. Lifetime value (BG/NBD + Gamma-Gamma, 182-day holdout)

Predicted holdout shopping days are 64,059 versus 63,212 actual (+1.3%); spend $2.30M versus $2.29M (+0.1%). Household-level MAE is 8.82 shopping days against 8.96 for a naive carried-forward rate: the model is right in aggregate and for ranking (Spearman 0.81), but only slightly better per household.

5. Coupons and campaigns (observational, no causal claim)

- 27 campaigns reached 1,559 households; 410 redeemed at least one coupon. Type A campaigns reach about 1,076 households each and have a 16.6% redeeming share of targeted, versus 6.5% for Type B.
- Households targeted in the test window churn less (naive gap -5.1 points), but within model risk tiers the gap is -0.1 points: targeting reached heavier shoppers (largest baseline standardised difference 1.42). There is no holdout, so the data cannot say whether campaigns work.

6. Value x risk targeting (ASSUMED ROI)

- 392 High value / Low risk households hold 45% of predicted CLV (observed churn 0.5%): protect with service, not discounts. Only 71 households are both high value and at elevated risk.
- Most risk sits in Low value households (710 High risk, observed churn 10.7%).
- **ASSUMED** 10% relative churn reduction, 25% margin, $2.00 per contact: the rule targets 116 households (5.0%) for expected net value $190 on $232 of cost; contacting everyone would be -$3,278. Across the ASSUMED sensitivity grid net value ranges from $0 to $4,545. These are scenarios, not forecasts.

7. Demographics (partial coverage)

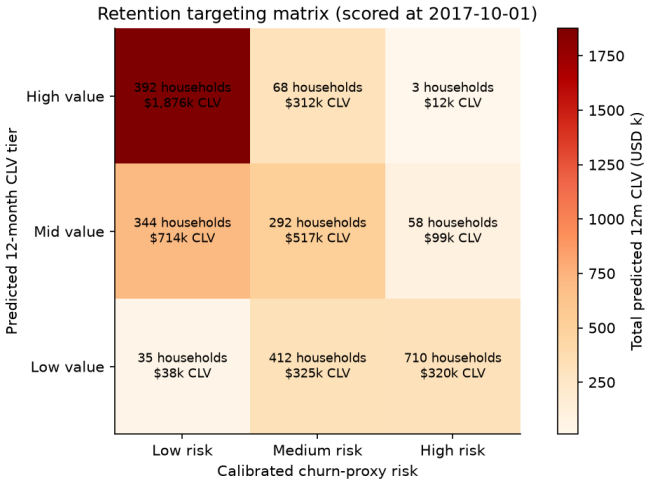
Only 801 of 2,469 households (32%) have demographics. They are heavier shoppers (median spend to date $1,981 versus $579) and only 2 of 801 scored covered households went quiet, so group churn rates are not informative.

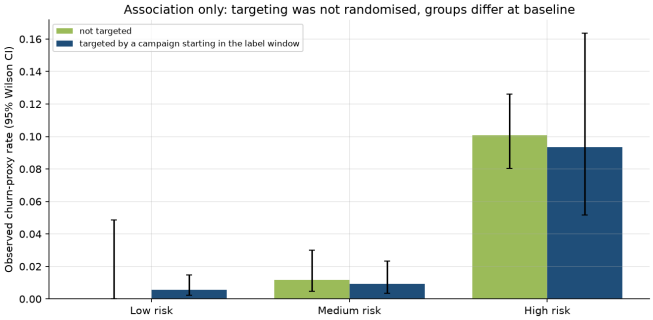
8. Recommendation and next step

Protect high-value households with service; do not spend discounts where risk is low. Before any ROI claim, run a randomised holdout campaign stratified by risk and value tier and replace the ASSUMED uplift with a measured one.

9. Limitations

One calendar year, one retailer, 2,469 loyalty-card households (card-less purchases are invisible); about 89 churn-proxy events per cutoff so model differences are within noise; the label is a proxy; partial demographics; BG/NBD has no seasonality and 12-month CLV extrapolates beyond one observed year; this extract contains no returns.


Retention targeting matrix (scored at 2017-10-01)
High value
Mid value
Low value
392 households
$1,876k CLV
68 households
$312k CLV
3 households
$12k CLV
344 households
$714k CLV
292 households
$517k CLV
58 households
$99k CLV
35 households
$38k CLV
412 households
$325k CLV
710 households
$320k CLV
Low risk
Medium risk
High risk
Calibrated churn-proxy risk
Predicted 12-month CLV tier
Total predicted 12m CLV (USD k)
1750
1500
1250
1000
750
500
250


Association only: targeting was not randomised, groups differ at baseline
not targeted
targeted by a campaign starting in the label window
Observed churn-proxy rate (95% Wilson CI)
0.16
0.14
0.12
0.10
0.08
0.06
0.04
0.02
0.00
Low risk
Medium risk
High risk